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MILITARY SPECIFICATION

CINNAMON NUT ROLL, CANNED

This limited coordination Military Specification has been prepared by the U.S. Army Natick Laboratories (GL), Natick, Mass. 01760, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-P-35000C. It is subject to modification. However, pending the promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers canned cinnamon nut roll for use by the Armed Forces as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-C-26 - Canned Baked, and Canned Steamed and Retorted Subsistence Items, Packaging and Packing Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.
MIL-STD-1105 - Sanitary Standards for Bakeries.

FSC 8920

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

U.S. Department of Agriculture

Regulations Governing the Grading and Inspection of Egg Products.

(Application for copies should be addressed to the U. S. Department of Agriculture, Poultry Division, Agricultural Marketing Service, Washington, D. C. 20250.)

U.S. Standards for Grades of Nonfat Dry Milk (Spray dried)
U.S. Standards for Grades of Shelled Almonds
U.S. Standards for Grades of Shelled Pecans

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Information Branch, Washington, D. C. 20250.)

Visual Aids for Inspection of Metal Containers

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

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American Oil Chemists' Society

Official and Tentative Methods.

(Application for copies should be addressed to the American Oil Chemists' Society, 508 S. 6th St., Champaign, Illinois 61820.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.1), the contractor shall furnish a sample for first article testing and approval (see 4.3.1, and 6.2).

3.2 Material. All material shall be clean, sound, wholesome, and free from evidence of insect infestation or other objectionable foreign matter, off-flavors or off-odors.

3.2.1 Sugar, granulated. Sugar shall be white, refined, granulated cane or beet sugar or a combination of both.

* 3.2.2 Flour. Flour shall be hard or soft wheat, bleached or unbleached, enriched. The flour shall be enriched in accordance with the Definitions and Standards of Identity for Enriched Flours as defined by the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

3.2.3 Yeast. Yeast shall be commercial compressed bakers' or active dry yeast.

3.2.4 Shortening. The shortening shall be hydrogenated cottonseed or peanut oil or a combination of both, and shall have a stability of not less than 100 hours active oxygen method (AOM).

* 3.2.5 Eggs, whole, fresh, frozen or dried. Egg products shall have been produced and handled in compliance with requirements set forth in the USDA publication entitled "Regulations Governing the Grading and Inspection of Egg Products". They shall be prepared under continuous inspection of the USDA and shall bear the inspection shield mark. Egg products shall have been laboratory tested and certified Salmonella negative by the USDA.

3.2.6 Milk, nonfat, dry. Nonfat dry milk shall meet the requirements for extra grade as defined in the U. S. Standards for Grades of Nonfat Dry Milk (spray dried).

3.2.7 Salt. Salt shall be white, refined sodium chloride. Iodized salt shall not be used

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* 3.2.8 Nuts. Nuts shall be either pecan or almond pieces. Pecan pieces shall be No. 1 pecan pieces of medium size, as defined in the U.S. Standards for Grades of Shelled Pecans. Almond pieces shall be No. 1 pieces, as defined in the U.S. Standards for Grades of Shelled Almonds. The almond pieces shall be commercially designated, medium size dice. Nuts shall be of the latest crop.

3.2.9 Flavoring. Flavoring shall be pure vanilla, vanillin, ethyl-vanillin, or any combination thereof.

3.2.10 Cinnamon. Cinnamon shall be the ground product obtained from the dried bark of cultivated varieties of Cinnamonum burmanni Blume or Cinnamomum zeylanicum, and shall contain not more than 6.0 percent ash, not more than 2.0 percent of ash insoluble in hydrochloric acid, and a volatile oil content of not less than 2 milliliters (ml) per 100 grams (g). It shall have a reddish-brown color.

3.2.10.1 Alternatively, Saigon cinnamon, Cassia, obtained from the dried cultivated varieties of Cinnamomum Loureirii Nees may be used, provided it complies with the requirements specified in 3.2.10, except that it shall contain a volatile oil content of not less than 3 ml per 100 g.

3.2.11 Cake crumbs. Cake crumbs shall be prepared from the following ingredients: Sugar, cake flour, hydrogenated shortening, whole eggs, nonfat dry milk, salt, and leavening agents. The formulation shall be so adjusted that the finished crumbled baked cake product, less all crust portion, shall contain not more than 20 percent moisture, not less than 10 percent fat, calculated on a moisture-free basis; and 100 percent shall pass through a U. S. No. 10 sieve.

3.3 Formulae.

3.3.1 Dough formula. The dough shall be made from ingredients in the following proportions, by weight:

Ingredients	Parts by weight
Flour	46.0
Nonfat dry milk	2.0
Shortening	16.0
Water	15.5
Yeast (compressed or equivalent of active dry yeast)	2.0
Sugar	10.0
Whole eggs (fresh, frozen or dried)	8.0
Salt	0.5

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3.3.1.1 Dough preparation. The roll dough shall be prepared by a straight dough method. The dry ingredients shall be sifted prior to mixing into the dough. All ingredients shall be mixed until the dough is thoroughly developed. The mixed dough shall be given sufficient floor time to achieve good sheeting qualities. (Precautionary note: Excessive floor time will result in over expansion of the dough during baking, creating fat separation and extrusion of filling from the cans during baking.) The mixed dough temperature shall not exceed 82°F., to facilitate handling and baking.

3.3.2 Filling formula. The filling shall be made from ingredients in the following proportions by weight:

Ingredients	Parts by weight
Sugar	23.00
Whole eggs (fresh, frozen or dried)	20.00
Shortening	14.00
Ground cinnamon	4.00
Salt	1.00
Flavoring	1.00
Cake crumbs	25.00
Pecan or almond pieces	12.00

3.3.2.1 Filling preparation. The dry ingredients, except the pecan or almond pieces, shall be sifted prior to mixing into the filling. All ingredients of the filling shall be mixed until a smooth, homogeneous paste is obtained. Mixing should be accomplished in such a manner as to secure a minimum of air incorporation. Prior to depositing, the filling shall be tempered (approximately 55° to 65°F.) to achieve optimum machinability.

3.3.3 Composition and makeup. The unbaked cinnamon nut roll shall consist of equal parts by weight of dough as specified in 3.3.1 and filling as specified in 3.3.2. The dough shall be sheeted mechanically (utilizing, preferably, double sheeting rolls) to a width of not less than 10 inches. In addition, the dough shall be uniformly sheeted so that the weight of 9 square inches shall not exceed 1 ounce. The filling shall be mechanically deposited on a sheet of uniform width and thickness so as to nearly cover the surface of the dough sheet. The dough sheeting and filling sheeting mechanisms shall be synchronized so as to yield equal weights of dough and filling over any given surface area. The resulting sheeted dough and filling shall then be mechanically and continuously rolled in a spiral (as a cinnamon bun) so that the finished roll shall have not less than 6 distinct layers of dough and 5 layers of filling when cut at right angles to the length of the roll. Cutting to the proper size and weight shall be accomplished mechanically, utilizing a guillotine or rotary miller cutter.

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3.4 Processing. The product shall be cut to size and deposited into the can (lined as specified in PPP-C-26) so that spirals show on top. The supplier shall deposit sufficient amount of the product to yield the net weight requirements specified in PPP-C-26. Immediately after the product has been deposited in the can, the lids of the cans shall be clinched on loosely to allow for escape of moisture and gases evolved during baking. It shall be baked in an oven at a temperature not to exceed 300°F. until thoroughly baked (see 3.5). The cans shall be hermetically sealed after being discharged from the oven. Sealing of the cans shall be timed so that the internal vacuum conforms to the requirements of PPP-C-26. By using a suitable cooling method, the internal temperature of the product, inside the cans, shall be reduced to below 100°F. before they are packed. Water immersion shall not be employed for cooling the cans.

* 3.5 Finished product. The canned product shall conform to the inside contour of the cans, except for the unfilled space above the top surface of the product. This unfilled space shall be not more than 5 percent of the interior volume of the can when examined at least 48 hours after processing. The baking time shall be adjusted so that the dough part of the nut roll is baked thoroughly and uniformly throughout. The finished baked product shall not possess a raw, stringy, or ungelatinized center when cut horizontally. The crust shall be browned but not burned, scorched, or possess an objectionable caramelized flavor. When the cooled nut roll is cut vertically and horizontally, it shall show dough and filling in nearly uniform proportion. The spiral in the horizontal cut shall have a uniform appearance. The filling shall not be runny or too dry. The finished product shall have a pleasant, cinnamon coffee cake flavor and odor. The product shall be free from mold.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.7 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-1105 or if produced in food establishments other than bakeries, shall meet the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preacceptance inspection (plant sanitation inspection). The product shall be rejected if produced in plants not meeting the requirements of 3.7.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 First article inspection (see 3.1, 6.1 and 6.2). The samples of the product which the contractor intends to supply shall be inspected to determine compliance with all product requirements of this specification. In the event these samples are found to be in noncompliance with these requirements, the Government reserves the right to disallow the contractor from commencing delivery production.

4.3.2 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2.1 Examination of ingredients. Examination shall be in accordance with 4.3.2.1.1 and 4.3.2.1.2. Nonconformance with identity or condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* 4.3.2.1.1 Identity. Examination shall be made of labels, invoices or other valid documents to determine conformance to identity requirements (kinds, grade, standards, source of inspection, Salmonella negative eggs, size, enrichment, etc.), as pertinent to the ingredient.

* 4.3.2.1.2 Condition. Each ingredient shall be examined organoleptically as necessary to determine conformance to condition requirements of 3.2. The sample unit for examination shall be approximately 1/2 pound of each ingredient (4 ounces of flavoring). The sample size shall be five. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot.

* 4.3.2.2 Testing of components. Testing of components described in this specification shall be as shown in table I. Results shall apply to the lot average. A test failure shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

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TABLE I. Testing of components (see 4.3.2)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & reqmt para
Shortening	One-pound composite 1/	Pounds	S-2	Fat stability	Pass or fail 2/	4.5 and 3.2.4
Cinnamon	8 oz. composite 1/	Ounces	S-2	Ash	Nearest 0.1%	4.5 and 3.2.10 or 3.2.10.1
				Volatile oil	Nearest ml/100 g	
Cake crumbs	One-pound composite 1/	Pounds	S-2	Fat	Nearest 1%	4.5 and 3.2.11
				Moisture	Nearest 1%	

1/ The composite shall be derived from the number of primary containers indicated by the inspection level.

2/ In the event of failure, report result to nearest hour.

4.3.3 Process examination. Examination shall be performed to determine conformance to formulation requirements for cake crumbs, dough and filling, and specific requirements pertinent to procedures, controls and machinery as contained in 3.3 and 3.4. Records of formulations and temperatures (as related to specific controls) shall be maintained. Nonconformance to one or more requirement(s), as determined by examination of records or by actual visual examination during processing, shall be cause for rejection of the involved product. In addition, process examination shall be performed in accordance with the examination criteria specified in PPP-C-26.

* 4.3.3.1 Cake crumbs. Cake crumbs shall be sieved to determine conformance to 3.2.11. The sample shall be a representative 50 gram composite of crumbs (less crust). Calculate results to nearest gram. Nonconformance shall indicate unacceptable cake crumbs and use of such shall be cause for rejection of the involved finished product.

* 4.3.4 Examination of end item. Examinations shall be in accordance with tables III through V and 4.3.4.1. The sampling plans for the aforementioned

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tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as percent defective for table III and defects per hundred units for tables IV and V.

TABLE II. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	AQLs Minor	AQLs Total
III	1 filled and sealed can	Cans	S-3	-	2.5	-
IV	1 filled and sealed can	Cans	S-2	-	4.0	-
V	Contents of 1 can	Cans	S-2	1.5	-	6.5

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Less than 107.7 grams (300 x 200 can)

1/ Report results to nearest 0.1 gram. Lot shall be rejected if sample data indicate a lot average net weight of less than 113.4 grams.

TABLE IV. Examination for fill of container

Category	Defect
Minor	
201	Unfilled space within the container, above the top surface of the product exceeds 5% of the can volume 48 hours or more after processing. 1/
202	Product touches top lid at point closer than 1/4 inch from side of can.

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TABLE IV. Examination for fill of container (cont'd)

Category	Defect
Minor (cont'd)	
203	Portion of product extends into the double seam of the can.
204	Product, except for top part, does not conform to the inside contour of can.

1/ Determination of unfilled volume above top surface of product. Determine the percentage of unfilled space within the container, above top surface of product, by the following procedure or one giving comparable results:

(a) Take an unopened can of product, fill the space between the lid and the top edge of the can with rape seed, and level off with a straight edge. Measure the volume of rape seed used to fill the space.

(b) Open the can. Fill any space at the side of the can with rape seed to the level of the top surface of the product. Do not measure this volume. From a measured quantity of rape seed, fill the remaining space to the top level of the can and level off with a straight edge. Return all seed falling over the edge of the can to the measured quantity and calculate the amount used.

(c) Empty the can, clean and fill it with rape seed, leveling off the top with a straight edge. Measure the volume.

(d) The percentage of unfilled space equals:

$$\frac{b \text{ minus } a}{c \text{ minus } a} \times 100$$

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TABLE V. Examination of product 1/ 2/

Category		Defect
Major	Minor	
101		Burnt or scorched.
102		Dough part of roll not baked thoroughly and uniformly throughout.
	201	Dough and filling not essentially uniform throughout.
	202	The spiral not uniform in appearance.
103		Filling runny or dry.
104		Not free of mold.
105		Not a pleasant cinnamon coffee cake flavor and odor.

1/ Prior to examining interior of roll, cut roll vertically and horizontally.

2/ Presence of foreign material (e.g., insects, insect parts, filth, wood, metal, glass, paint, etc.), shall be cause for rejection of the lot.

* 4.3.4.1 Examination of packaging, packing, marking and labeling. Examinations shall be in accordance with all the inspection criteria for the end item as contained in PPP-C-26, except that the examination for net weight shall be in accordance with table III of this specification, and USDA Visual Aids for Inspection of Metal Containers shall be used as a guide (illustrations only) in classifying external can defects.

4.5 Tests. Test procedures and controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide as a minimum, such quality assurance, that supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

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4.5.1 Examination procedure.

4.5.1.1 Sieving procedure. Sieving shall be performed as follows to determine the percentage of cake crumbs that will pass through a U.S. Standard Woven-Wire-Cloth sieve: Place 50 grams of the cake crumbs on an 8-inch full height U.S. Standard No. 10 sieve to which the bottom pan has been attached. Place the assembly on a smooth level surface and with a steady sieving motion, move it 10 inches to the left and return to its original position, repeating the movement 20 times in approximately 13 seconds. Calculate the percentage by multiplying the weight of the material in the colleting pan by two (2).

4.5.2 Test procedure.

4.5.2.1 Chemical analyses.

4.5.2.2 Cinnamon. Cinnamon shall be tested for total ash, ash insoluble in HCl and volatile oil in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Spices and Other Condiments, section: Spices:

Test	Method
Total ash	Ash
Ash insoluble in HCl	Ash Insoluble in Acid
Volatile oil	Volatile and Nonvolatile Ether Extract

4.5.2.3 Cake crumbs. Cake crumbs shall be tested for moisture and fat contents in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists:

Test		Chapter and Section	Method
Moisture	Chapter:	Sugars and Sugar Products	Moisture (Vacuum drying)
	Section:	Sugars, Sirups and Molasses	
Fat	Chapter:	Cereal Foods	Fat - Official
	Section:	Baked Products Other than Bread	

4.5.2.4 Hydrogenated shortening. Hydrogenated shortening shall be tested for fat stability in accordance with Official and Tentative Methods of the American Oil Chemists' Society, 2nd Edition (1946), section C, Commercial Fats and Oils, method: Cd 12-57.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and labeling. The product shall be packaged, packed and labeled in accordance with PPP-C-26, as specified for pecan cake roll, canned, except that the product shall be labeled "Cinnamon Nut Roll, Canned".

5.2 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3.1, and 6.2).

* 6.2 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of 6 - 300 x 200 cans. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and the relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

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